

Phil Piwonka

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Principal Software Engineer with more than a decade building and operating foundational AWS infrastructure across Lambda and Bedrock.

WORK

Principal Software Development Engineer

Sep 2025 - Present

AWS Bedrock

Recruited to Project Mantle during its early expansion, helping evolve the inference engine that became the backbone of Amazon Bedrock.

- Partner with OpenAI to align Amazon Bedrock's Responses API behavior with OpenAI's first-party implementation.
- Built and launched server-side tool execution through AWS Lambda and Amazon Bedrock AgentCore Gateway connectors, allowing customers to use existing MCP-compatible tools during inference.
- Built and launched Web Search for OpenAI frontier models by integrating Amazon search infrastructure with IAM authorization, usage limits, fairness controls, and production monitoring while keeping customer queries within the AWS environment.

Principal Software Development Engineer

Sep 2019 - Sep 2025

AWS Lambda

Senior Software Development Engineer

Apr 2016 - Sep 2019

Led the architecture and evolution of Lambda's placement and capacity-management systems as the service expanded across new virtualization platforms, processors, storage configurations, and workload sizes.

- Designed and built Lambda's placement service and engine to manage oversubscription, thermal constraints, and efficient workload placement across the compute fleet.
- Built simulators and utilization models that informed workload entitlements, fleet planning, and placement strategy; partnered with EC2, finance, and capacity-planning teams to improve warm capacity and fleet flexibility.
- Drove Lambda's multi-year migration to Firecracker, rewriting capacity-management and tenancy architecture while the service continued to grow; co-authored the NSDI '20 Firecracker paper.
- Expanded capacity systems across Intel, AMD, and AWS Graviton processors; NVMe- and EBS-backed hosts; and Lambda environments ranging from 1.5 GB to 10 GB.
- Launched Lambda support for container images up to 10 GiB; co-authored the resulting USENIX ATC '23 Best Paper describing a system that has processed hundreds of trillions of invocations for more than one million customers.
- Launched Lambda SnapStart to reduce cold starts for latency-sensitive applications.
- Served as a senior operational reviewer across Lambda, ECS Fargate, and App Runner and joined Lambda's escalation on-call rotation based on broad system and incident-response expertise.

Senior Software Development Engineer

Nov 2014 - Apr 2016

Amazon Kindle

Software Development Engineer II

Sep 2010 - Nov 2014

Built customer-facing software and backend services across the original Kindle Fire, cross-platform reading applications, and Goodreads integration.

- Co-developed ACX, an HTML5 framework used to ship Wikipedia Lookup, passage translation, X-Ray, and End Actions across desktop, Android, and iOS Kindle applications.
- Built the original Kindle Fire home-screen launcher and carousel, using custom GPU-accelerated rendering to maintain responsive scrolling across books, music, and video.
- Built the backend integration for Goodreads on Kindle, moving its Ruby on Rails stack into Amazon infrastructure and adding caching and protective services for e-reader and Fire tablet traffic.
- Helped decompose Goodreads' Rails monolith into services and launched Goodreads Deals, a personalized ebook campaign whose batch pipeline was isolated from online customer traffic.

Software Development Engineer

Nov 2006 - Sep 2010

Microsoft

- Developed database providers for Microsoft Sync Framework and integrated them into Visual Studio 2008 and 2010, enabling peer-to-peer synchronization across applications, services, and devices.
- Designed Merge Replication features for SQL Server 2008 and supported enterprise customers through urgent production issues.

Software Engineer

Jun 2003 - Oct 2006

Lockheed Martin

- Designed and validated mission-critical numerical and clustering algorithms in a near-real-time, high-performance UNIX parallel-computing environment.
- Improved algorithm performance with multithreading and shared-memory programming using C++, MATLAB, and Perl; led delivery work across multiple customer organizations.

PUBLICATIONS

Firecracker: Lightweight Virtualization for Serverless Applications

USENIX Association · 2020

Co-author. Describes the design of Firecracker and AWS Lambda's migration to the new virtualization platform, which supports millions of production workloads and trillions of requests per month.

On-demand Container Loading in AWS Lambda

USENIX Association · Jul 2023

Co-author and USENIX ATC '23 Best Paper. Describes Lambda's storage and caching system for container images, which has processed hundreds of trillions of invocations for more than one million customers.

EDUCATION

University of Colorado

B.S. and M.S. · Computer Science; minor in Applied Mathematics

Jun 1998 - Jun 2003